

## OS SKILL 7

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    pid_t pid;
    int status;

    printf("=== PATH Variable ===\n");
    char *path = getenv("PATH");
    if (path != NULL) {
        printf("PATH = %s\n", path);
    }

    printf("\n=== PATH Directories ===\n");
    if (path != NULL) {
        char copy[4096];
        strncpy(copy, path, sizeof(copy) - 1);
        copy[sizeof(copy) - 1] = '\0';

        char *dir = strtok(copy, ":");
        while (dir != NULL) {
            printf("%s\n", dir);
            dir = strtok(NULL, ":");
        }
    }

    printf("\n=== Locate Executables ===\n");
    system("which ls");
    system("which gcc");
    system("which bash");

    printf("\n=== Verify Permissions ===\n");
    system("ls -l /bin/ls");

    printf("\n=== Child Process and waitpid() ===\n");

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }
}

```

```

    if (pid == 0) {
        printf("Child process started. PID = %d\n", getpid());

        execlp("ls", "ls", "-l", NULL);

        perror("execlp failed");
        exit(1);
    } else {
        printf("Parent process waiting for child PID = %d\n", pid);

        waitpid(pid, &status, 0);

        if (WIFEXITED(status)) {
            printf("Child exited with status = %d\n", WEXITSTATUS(status));
        } else {
            printf("Child did not terminate normally\n");
        }
    }

    printf("\n=== Missing Command Test ===\n");

    pid = fork();

    if (pid == 0) {
        execlp("wrongcommand", "wrongcommand", NULL);
        perror("Command not found");
        exit(1);
    } else {
        waitpid(pid, &status, 0);

        if (WIFEXITED(status)) {
            printf("Missing command child exited with status = %d\n",
                WEXITSTATUS(status));
        }
    }

    printf("\n=== Command Resolution Test ===\n");
    system("command -v ls");
    system("command -v bash");
    system("command -v wrongcommand");

    return 0;
}

```

```

=== PATH Variable ===
PATH = /usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_3.0.1.0_x64_8w
yb3d8bbwe:/mnt/c/Python314/Scripts:/mnt/c/Python314:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common File
Oracle/Java/javapath:/mnt/c/Windows/System32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files/nodejs:/mnt/c/Pro
gramData/chocolatey/bin:/mnt/c/Program Files/MSI:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/Geetha/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/Geetha/AppData/Local/Programs/Microsoft VS C
o/bin:/mnt/c/Users/Geetha/AppData/Roaming/npm:/snap/bin

=== PATH Directories ===
/usr/local/sbin
/usr/local/bin
/usr/sbin
/usr/bin
/sbin
/bin
/usr/games
/usr/local/games
/usr/lib/wsl/lib
/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_3.0.1.0_x64_8wkyb3d8bbwe
/mnt/c/Python314/Scripts/
/mnt/c/Python314/
/mnt/c/Program Files/Common Files/Oracle/Java/javapath
/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath
/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath
/mnt/c/Windows/System32
/mnt/c/Windows
/mnt/c/Windows/System32/Wbem
/mnt/c/Windows/System32/WindowsPowerShell/v1.0/
/mnt/c/Windows/System32/OpenSSH/
/mnt/c/Program Files/nodejs/
/mnt/c/ProgramData/chocolatey/bin
/mnt/c/Program Files/MSI/
/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin/
/mnt/c/Users/Geetha/AppData/Local/Microsoft/WindowsApps
/mnt/c/Users/Geetha/AppData/Local/Programs/Microsoft VS Code/bin
/mnt/c/Users/Geetha/AppData/Roaming/npm
/snap/bin

```

=== Locate Executables ===

```

/usr/bin/ls
/usr/bin/gcc
/usr/bin/bash

```

=== Verify Permissions ===

```
lrwxrwxrwx 1 root root 29 Mar 30 2026 /bin/ls -> ../lib/cargo/bin/coreutils/ls
```

=== Child Process and waitpid() ===

Parent process waiting for child PID = 526

Child process started. PID = 526

total 24

```
-rwxr-xr-x 1 nishanth nishanth 16480 Oct  1 13:04 waitpid
```

```
-rw-r--r-- 1 nishanth nishanth 2059 Oct  1 13:04 waitpid.c
```

Child exited with status = 0

=== Missing Command Test ===

Command not found: No such file or directory

Missing command child exited with status = 1

=== Command Resolution Test ===

```
/usr/bin/ls
```

```
/usr/bin/bash
```